



BLUESTOCK MUTUAL FUND ANALYTICS

Capstone Project Report

Field	Details
Prepared by	Tadepalli Haneesh
Role	Data Analyst Intern
Teams ID	59FMBF
Organization	Bluestock Fintech
Submission Date	12th June 2026
Email	haneeshtadepalli@gmail.com
GitHub	https://github.com/hanicode2025/bluestock_mf_capstone

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1. Executive Summary

This report presents a comprehensive analysis of 40 Indian mutual fund schemes across 10 fund houses, covering the period January 2022 to June 2026. The project was developed as part of the Bluestock Fintech Data Analyst Internship Capstone.

Key Findings

- Analysed 64,320 NAV records across 40 schemes after cleaning and forward-filling weekends/holidays
- Mirae Asset Large Cap Fund has the highest Sharpe Ratio (1.07), indicating best risk-adjusted returns
- ICICI Pru Midcap Fund topped the composite fund scorecard with a perfect score of 100/100
- SBI Small Cap Fund experienced the worst maximum drawdown of -52.6% in October 2025
- Only 2.2% of SIP investors maintain regular payment intervals — 97.8% are classified as at-risk
- Industry AUM peaked at approximately Rs. 15 lakh crore in 2024 before correcting in 2025
- Total of 32,778 investor transactions analysed across 11 states

Deliverables Completed

- ETL pipeline and data ingestion scripts (Python)
- SQLite database with star schema (6 tables, 97,000+ rows)
- EDA notebook with 15+ charts and 10 key insights
- Performance analytics with CAGR, Sharpe Ratio, Beta, VaR, Max Drawdown
- Interactive Tableau dashboard with 4 pages and slicers
- Advanced analytics with VaR/CVaR, cohort analysis, and fund recommender system
- Final report and 12-slide presentation

2. Data Sources

2.1 AMFI Provided Datasets

The following 10 CSV datasets were provided by Bluestock Fintech for analysis:

Dataset	Rows	Description
01_fund_master	40	Fund metadata, categories, risk grades, expense ratios
02_nav_history	46,000	Daily NAV for all 40 schemes (2022-2026)
03_aum_by_fund_house	90	Monthly AUM by fund house
04_monthly_sip_inflows	48	Monthly SIP inflow data with YoY growth
05_category_inflows	144	Net inflows by fund category
06_industry_folio_count	21	Industry folio count trends by asset class
07_scheme_performance	40	Pre-computed performance metrics per scheme
08_investor_transactions	32,778	Individual investor transaction records
09_portfolio_holdings	322	Stock holdings per fund with weights
10_benchmark_indices	8,050	Nifty 50 and Nifty 100 daily closing values

2.2 Live NAV Data (mfapi.in API)

Live NAV data was fetched for 5 large cap schemes using the mfapi.in REST API:

- SBI Large Cap Fund (AMFI Code: 119598) — 3,307 rows
- ICICI Prudential Large Cap Fund (AMFI Code: 120586) — 3,298 rows
- Nippon India Large Cap Fund (AMFI Code: 118632) — 3,298 rows
- Axis Large Cap Fund (AMFI Code: 120465) — 3,307 rows
- Mirae Asset Large & Midcap Fund (AMFI Code: 118834) — 3,298 rows

2.3 Data Quality Summary

- Only 1 anomaly found: 12 null values in yoy_growth_pct column (expected — first 12 months have no prior year for comparison)
- All 40 AMFI codes in fund_master validated against nav_history — 100% match

- NAV history expanded from 46,000 to 64,320 rows after forward-filling weekends and market holidays
- No duplicate records found across any dataset
- All date columns standardised to datetime format

3. ETL Design & Architecture

3.1 Pipeline Overview

The ETL pipeline consists of three stages: Extract, Transform, and Load. All steps are implemented in Python using Pandas, SQLAlchemy, and the Requests library.

Stage	Tool	Output
Extract	requests, pandas.read_csv	10 raw CSVs + 5 live NAV CSVs in data/raw/
Transform	pandas, numpy	10 cleaned CSVs in data/processed/
Load	SQLAlchemy, SQLite	bluestock_mf.db with 6 tables

3.2 Data Cleaning Steps

- nav_history: Parsed dates to datetime, sorted by amfi_code + date, forward-filled missing NAV for weekends and holidays, removed duplicates, validated NAV > 0
- investor_transactions: Standardised transaction_type to Title Case, validated amount > 0, fixed date formats, verified KYC status enum values
- scheme_performance: Validated all return values are numeric, checked expense_ratio range (0.1% to 2.5%), found zero anomalies
- All other datasets: Parsed date/month columns, filled yoy_growth_pct nulls with 0, removed duplicates

3.3 SQLite Star Schema

The database follows a star schema design with 2 dimension tables and 4 fact tables:

- dim_fund: 40 rows — fund metadata, risk grades, expense ratios
- dim_date: 1,608 rows — full date dimension with year, month, quarter, weekend flag
- fact_nav: 64,320 rows — daily NAV per scheme
- fact_transactions: 32,778 rows — investor transactions
- fact_performance: 40 rows — scheme performance metrics
- fact_aum: 90 rows — monthly AUM by fund house

4. EDA Findings

4.1 Fund Universe Overview

- 10 fund houses analysed: SBI, HDFC, ICICI Prudential, Nippon India, Kotak, Axis, ABSL, UTI, Mirae Asset, DSP
- 2 categories: Equity (28 schemes) and Debt (12 schemes)
- 12 sub-categories including Large Cap, Mid Cap, Small Cap, Flexi Cap, ELSS, Gilt, Liquid
- 5 risk grades: Low, Moderate, Moderately High, High, Very High

4.2 NAV Trends

- All equity funds showed strong recovery in 2023 after the 2022 correction
- Mid Cap and Small Cap funds significantly outperformed Large Cap in 2023-2024
- 2025 witnessed a broad market correction, with Small Cap funds hit hardest
- Liquid and Gilt funds showed stable, low-volatility NAV growth throughout

4.3 SIP & AUM Trends

- Industry SIP inflows grew consistently from Rs. 11,517 Cr (Jan 2022) to Rs. 31,002 Cr (Dec 2025) — a 169% increase
- Active SIP accounts grew from 4.91 Cr to over 10 Cr during the period
- Total industry folio count grew from 13.26 Cr to 26.12 Cr — doubling in 4 years
- SBI Mutual Fund leads AUM with Rs. 6.05 lakh crore as of 2022, maintaining dominance throughout

4.4 Investor Demographics

- Maharashtra and Uttar Pradesh lead in total transaction amounts
- Age group 26-35 contributes the highest SIP volumes
- T30 cities account for approximately 65% of total investment amounts
- UPI and Net Banking are the most popular payment modes

5. Performance Analysis

5.1 CAGR Comparison

5-year CAGR was computed as: $CAGR = (NAV_end / NAV_start)^{(1/n_years)} - 1$, using 365.25 calendar days for annualisation.

Fund	1yr CAGR	3yr CAGR	5yr CAGR
SBI Small Cap Regular	82.9%	26.7%	32.4%
DSP Small Cap Regular	65.2%	27.0%	32.3%
HDFC Mid-Cap Opportunities Regular	53.3%	32.4%	30.1%
ABSL Frontline Equity Regular	47.9%	29.0%	23.5%
Mirae Asset Large Cap Regular	20.4%	34.0%	30.9%

5.2 Risk-Adjusted Returns

Sharpe Ratio = $(R_p - R_f) / \text{Std}(R_p) \times \sqrt{252}$, using $R_f = 6.5\%$ (RBI repo rate proxy)

Fund	Sharpe Ratio	Sortino Ratio
Mirae Asset Large Cap Regular	1.07	1.49
Kotak Flexicap Regular	0.97	1.48
Mirae Asset Tax Saver Regular	0.92	1.35
ICICI Pru Midcap Regular	0.88	1.29
SBI Bluechip Regular	0.86	1.29

5.3 Alpha & Beta

Alpha and Beta were computed using OLS regression of fund daily returns against Nifty 100 returns.

- SBI Small Cap Regular showed highest alpha (0.303), indicating strong outperformance over benchmark
- R-squared values were very low across all funds, confirming active management independence from benchmark
- Most equity funds had beta close to 0, suggesting low correlation with Nifty 100 in this dataset

5.4 Maximum Drawdown

- Worst drawdown: SBI Small Cap Direct at -52.6% (October 2025)
- Axis Small Cap: -51.7% (May 2026)
- Best performers: ICICI Pru Liquid (-0.10%), Kotak Liquid (-0.12%)
- Large cap funds averaged -15% to -18% max drawdown

5.5 Fund Scorecard

Composite score (0-100) = 30% x 3yr return rank + 25% x Sharpe rank + 20% x Alpha rank + 15% x expense ratio rank (inverse) + 10% x max drawdown rank (inverse)

Rank	Fund	Score
1	ICICI Pru Midcap Regular	100.0
2	Axis Midcap Regular	96.3
3	HDFC Mid-Cap Opportunities Regular	94.6
4	Mirae Asset Large Cap Regular	94.0
5	Kotak Flexicap Regular	91.9

6. Advanced Analytics

6.1 Value at Risk (VaR) & CVaR

Historical VaR (95%) = 5th percentile of daily return distribution. CVaR = mean of returns below VaR threshold.

- Small Cap funds carry highest risk: ABSL Small Cap VaR = -2.39%, Axis Small Cap VaR = -2.33%
- Large Cap funds average VaR around -1.2% to -1.3%
- Liquid funds have near-zero VaR (<0.02%), confirming capital preservation role
- CVaR consistently 25-35% worse than VaR, indicating fat tail risk in equity funds

6.2 Rolling 90-Day Sharpe Ratio

- All 5 large cap funds showed negative Sharpe ratios during the 2022 market correction
- Strong recovery in Sharpe ratios observed from mid-2023 onwards
- Mirae Asset Large Cap consistently maintained the highest rolling Sharpe ratio
- All funds showed elevated volatility in late 2025 correction

6.3 Investor Cohort Analysis

Cohort Year	Investors	Avg SIP Amount (Rs.)	Total Invested (Rs. Cr)	Top Fund
2024	4,803	1,07,423	349.1	UTI Nifty 50 Index Fund
2025	197	1,09,159	30.5	SBI Small Cap Direct

6.4 SIP Continuity Analysis

- 1,362 investors had 6 or more SIP transactions — eligible for continuity analysis
- Only 30 investors (2.2%) maintained regular SIP intervals of 35 days or less
- 1,332 investors (97.8%) flagged as at-risk with avg gaps exceeding 35 days
- Average gap for at-risk investors ranged from 40 to 93 days
- This indicates a significant SIP discontinuation problem requiring intervention

6.5 Fund Recommender System

A rule-based recommender was built in recommender.py, mapping risk appetite to fund risk grades and ranking by Sharpe ratio:

- Low Risk: ICICI Pru Liquid (Sharpe: 7.68), Kotak Liquid (6.18), ABSL Liquid (5.14)

- Moderate Risk: HDFC Top 100 Regular (1.06), Mirae Asset Large Cap (1.06), ICICI Pru Bluechip Direct (1.03)
- High Risk: Kotak Emerging Equity (0.96), ICICI Pru Midcap (0.95), SBI Small Cap Regular (0.94)

6.6 Sector HHI Concentration

- Axis Bluechip Regular has highest HHI (0.206) with only 10 stocks — most concentrated
- SBI Small Cap Regular has lowest HHI (0.107) with 12 stocks — most diversified
- Funds with fewer holdings (8-9 stocks) tend to have higher concentration risk

7. Dashboard Overview

An interactive Tableau dashboard was built with 4 pages, all charts include tooltips, slicers, and the Bluestock branding.

Page	Charts	Slicers
Page 1 - Industry Overview	KPI cards (4), AUM bar chart, AUM trend line	Year, Category
Page 2 - Fund Performance	Scatter plot (return vs risk), Fund scorecard table, NAV vs benchmark	Fund House, Category, Plan
Page 3 - Investor Analytics	Transaction by state bar, Pie chart (SIP/Lumpsum/Redemption), Age group bar, Monthly volume line	State, Age Group, City Tier
Page 4 - SIP & Market Trends	SIP vs Nifty50 dual axis line, Category inflow heatmap, Top 5 categories bar	Month, YoY Growth

- Dashboard file: dashboard/bluestock_mf_dashboard.twbx
- Drill-through action configured from Fund Performance page to NAV detail
- Bluestock logo added to all 4 dashboard pages

8. Limitations

- The mfapi.in API returned incorrect scheme names for some AMFI codes — 5 alternative confirmed funds were used instead of the originally specified codes
- The dataset covers only 40 schemes out of 1,908+ schemes in the Indian mutual fund industry — findings may not be fully representative
- Investor transaction data covers only 2024-2025, limiting long-term cohort analysis
- AUM data has the same values repeated across dates, limiting time-series AUM analysis
- Portfolio holdings data is available only for a single date (December 2025), preventing historical sector analysis
- All calculations use historical data and should not be treated as investment advice

9. Recommendations

For Investors

- Investors with moderate risk appetite should consider Mirae Asset Large Cap and Kotak Flexicap based on superior risk-adjusted returns
- Small Cap fund investors should be aware of high VaR (-2.3% to -2.4%) and max drawdown (-52%) risks
- SIP investors should maintain consistent monthly intervals to maximise rupee cost averaging benefits

For Fund Houses

- The 97.8% SIP at-risk rate signals an urgent need for automated SIP reminder and continuation systems
- Funds with high HHI (above 0.18) should consider increasing diversification to reduce concentration risk
- The 2025 correction highlighted vulnerability of Small Cap funds — clearer risk communication to investors is recommended

For Bluestock Platform

- Integrate the fund recommender system into the platform for personalised fund suggestions based on risk appetite
- Add real-time NAV alerts for investors when their fund drops below a defined threshold
- Build automated weekly performance reports using the HTML email generator (Bonus B5)

10. Conclusion

This capstone project successfully delivered a complete end-to-end mutual fund analytics solution for Bluestock Fintech. Starting from raw CSV data and live API feeds, the project built a robust ETL pipeline, a well-structured SQLite database, comprehensive exploratory and performance analytics, and an interactive Tableau dashboard.

Key takeaways from the analysis include the consistent outperformance of mid cap and flexi cap funds on risk-adjusted metrics, the alarming SIP discontinuation rate among investors, and the potential of data-driven fund recommendation systems to guide investor decisions.

All code, notebooks, datasets, and deliverables have been version-controlled on GitHub and are available at:

https://github.com/hanicode2025/bluestock_mf_capstone